Riemannian Conjugate Gradient Method for 1D Burgers Enstrophy Maximization

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Contents

1 1D Burgers Equation

1.1 Formulation

The global existence and uniqueness of classical solutions to the Navier-Stokes system of equations, with respect to time, for any sufficiently regular initial data, is the foremost problem in mathematical fluid mechanics [6][21]. The Burgers equation is a fluid dynamics model that is used as a "toy" model towards studying the Navier-Stokes system [24]. This is due to the Burgers equation's presence of a quadratic first-order term and diffusion term, with the noted simplification relative to the Navier-Stokes system being the absence of a pressure term in the mathematical equation [10]. The one-dimensional viscous Burgers equation (1DB) can be expressed for velocity ϕ in space x and time t as

$$\begin{cases} \partial_t \phi + \frac{1}{2}\partial_x \phi^2 - \nu \partial_{xx}\phi = 0 & \in (0,T] \times \Omega \\ \phi(0,x) = \varphi \end{cases} \tag{1}$$

for periodic domain $\Omega = \mathbb{T}$ and initial condition φ.

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1.2 Finite-Time Enstrophy Growth

One important quantity characterizing solutions to the aforementioned systems is defined as enstrophy $\mathcal{E}$ [19], which can be defined as "cumulative energy dissipation" [23] and expressed for the Burgers equation as

$$\mathcal{E}(\phi(t)) = \frac{1}{2}\int_{\Omega} |\partial_x \phi(t,x)|^2 \, dx \ . \tag{2}$$

Moreover, boundedness of $\mathcal{E}(\phi(t))$ over a finite-time interval $[0, T]$ guarantees that $\phi(t)$ is a smooth solution to the system, and this can be determined by the maximum enstrophy value via inspection of $d\mathcal{E}/dt$. The question of finite-time boundedness of enstrophy for the Burgers equation under these "extreme" [9] conditions (maximizing enstrophy) has recently become a closed problem [2], and the finite-time maximum enstrophy for initial enstrophy value $\mathcal{E}_0$ with respect to initial data φ is known to adhere to the following two power laws [4]:

$$\max_T \left(\max_{\varphi} \ \mathcal{E}(T) \right) \approx K_1 \mathcal{E}_0^{3/2} \quad , \quad K_1 > 0 \ ; \tag{3}$$

$$\operatorname*{argmax}_T \left(\max_{\varphi} \ \mathcal{E}(T) \right) \approx \frac{K_2}{\sqrt{\mathcal{E}_0}} \quad , \quad K_2 > 0 \ . \tag{4}$$

This is the behavior of solutions to optimization problems of the form

$$\begin{aligned} \widetilde{\varphi} &= \operatorname*{argmax}_{\varphi \in H^1(\Omega)} \mathcal{J}_T(\varphi) \\ \text{given } \mathcal{E}_0 &= \frac{1}{2} \left\| \frac{d\varphi}{dx} \right\|_{L_2}^2 \end{aligned} \tag{5}$$

where the objective functional $\mathcal{J}(\phi)$ can be suitably defined as

$$\mathcal{J}_T(\varphi) = \int_0^T \frac{d}{d\tau} \mathcal{E}(\tau;\varphi) \ d\tau = \mathcal{E}(T;\varphi) - \mathcal{E}_0 \ , \tag{6}$$

which represents the finite-time enstrophy growth for initial enstrophy value $\mathcal{E}_0$ and time window T.

2 PDE-Constrained Optimization

2.1 Gradient Descent

Gradient Descent upholds the "greedy" principle in mathematical optimization of finding the locally optimal choice of direction when searching for a global optimal solution [14][25][16]. It is a first-order discrete optimization method since it relies on the gradient of an objective functional $\nabla \mathcal{J}(\phi)$, in contrast to second-order Newton methods that require computation of the Hession of an objective functional $\mathcal{H}(\mathcal{J}(\phi))$.

In the continuous setting, where the analogous idea of the gradient method is referred to as "Gradient Flow", the dynamical system for data ϕ in pseudo-time τ can be represented as a first-order differential equation

$$\begin{cases} \frac{d\phi}{d\tau} = -\nabla\mathcal{J}(\phi) \\ \phi(0) = \phi_0 \end{cases} , \tag{7}$$

where $\nabla\mathcal{J}(\phi) = 0$ guarantees a locally optimal solution in the nonconvex setting. Back in the discrete setting for a nonlinear problem, the differential equation in (7) can be expressed by a finite-difference approximation for current data $\phi^{(n)}$ and step-size $\tau^{(n)}$ as

$$\frac{\phi^{(n+1)} - \phi^{(n)}}{\tau^{(n)}} = -\nabla\mathcal{J}(\phi^{(n)}) \ . \tag{8}$$

Equivalently, (8) is the explicit Euler method [11] for solving initial-value problems:

$$\phi^{(n+1)} = \phi^{(n)} - \tau^{(n)}\nabla\mathcal{J}(\phi^{(n)}) \ . \tag{9}$$

For gradient-based optimization, (9) is called the "Steepest Descent" method. The Steepest Descent method takes orthogonal steps toward the optimal solution based on the initial search direction, which can result in a slow convergence rate.

2.2 Conjugate Gradient

The well-known "Conjugate Gradient" method [8] improves upon the gradient descent by resetting the search direction after each step to proceed towards the optimal solution:

$$\begin{cases} \phi^{(n+1)} = \phi^{(n)} + \tau^{(n)} d^{(n)} \\ d^{(n)} = -\nabla\mathcal{J}(\phi^{(n)}) + \beta^{(n)} d^{(n-1)} \end{cases} . \tag{10}$$

In (10), the step-size $\tau^{(n)}$ can be chosen as the optimal length in the direction of $d^{(n)}$, and the scalar "momentum" parameter $\beta^{(n)}$ can be computed for the nonlinear problem using the Polak-Ribière formula:

$$\beta_n = \frac{\langle \nabla\mathcal{J}^{(n)}, \nabla\mathcal{J}^{(n)} - \nabla\mathcal{J}^{(n-1)} \rangle}{\|\nabla\mathcal{J}^{(n-1)}\|^2} \ . \tag{11}$$

The downside of the Conjugate Gradient approach relative to Steepest Descent might occur in the case that the geometry of the problem makes finding each new optimal search direction an expensive process. Nevertheless, given a suitable geometric setting, the Conjugate Gradient method is superior to the Steepest Descent method.

2.3 Momentum-Based Gradient Descent

In comparison to system (7), consider a second-order differential equation where $\nabla\mathcal{J}(\phi) = 0$ guarantees a locally optimal solution:

$$\begin{cases} \nu_1 \frac{d^2\phi}{d\tau^2} = -\nabla\mathcal{J}(\phi) - \nu_2 \frac{d\phi}{d\tau} \\ \phi(0) = \phi_0 \end{cases} . \tag{12}$$

In the physical sense of this system, ν_1 and ν_2 can be respectively viewed to represent the "mass" and "friction/viscosity" of the system so that the trajectory of the gradient is influenced by so-called "momentum" [25]. Just as how (9) is a discretization of (7), the differential equation in (12) can be expressed by a finite-difference approximation for current data $\phi^{(n)}$ and step-size $\tau^{(n)}$ as:

$$\nu_1 \frac{\phi^{(n+1)} - 2\phi^{(n)} + \phi^{(n-1)}}{(\tau^{(n)})^2} = -\nabla\mathcal{J}(\phi^{(n)}) - \nu_2 \frac{\phi^{(n)} - \phi^{(n-1)}}{\tau^{(n)}} \tag{13}$$

$$\Leftrightarrow \phi^{(n+1)} = \phi^{(n)} - \frac{(\tau^{(n)})^2}{\nu_1}\nabla\mathcal{J}(\phi^{(n)}) + \left(1 - \frac{\nu_2}{\nu_1}\right)\tau^{(n)}\left(\phi^{(n)} - \phi^{(n-1)}\right) . \tag{14}$$

Defining α and β in terms of $\nu_1, \nu_2, \tau^{(n)}$ from (14) results in the following momentum-based version of gradient descent called the "Heavy-Ball" method [15]:

$$\phi^{(n+1)} = \phi^{(n)} - \alpha\nabla\mathcal{J}(\phi^{(n)}) + \beta(\phi^{(n)} - \phi^{(n-1)}) . \tag{15}$$

By defining $d^{(n)} = \phi^{(n+1)} - \phi^{(n)}$, which implies

$$d^{(n)} = -\alpha\nabla\mathcal{J}(\phi^{(n)}) + \beta(d^{(n-1)}) , \tag{16}$$

the Heavy-Ball method can be rewritten as

$$\begin{cases} \phi^{(n+1)} = \phi^{(n)} + d^{(n)} \\ d^{(n)} = -\alpha\nabla\mathcal{J}(\phi^{(n)}) + \beta d^{(n-1)} \end{cases} . \tag{17}$$

It is clear that (17) is similar (10), and in fact they have the same convergence rate asymptotically for convex quadratic functions [16]. In fact, the Conjugate Gradient method may be referred to as an "Optimal Heavy-Ball" method [22].

3 Numerical Approach

3.1 Riemannian Optimization

Consider that a PDE constraint φ belongs to a Riemannian manifold $\mathcal{M}$. Equivalently,

$$\varphi \in \mathcal{M}_K \tag{18}$$

Any gradient optimization algorithm such as (9), (10), (17) does not guarantee that the iterative update will belong to $\mathcal{M}_K$. The "Riemannian Conjugate Gradient" method [1][5] enforces the PDE constraint on the gradient optimization algorithm by ensuring that the new chosen direction is along a subspace tangent to $\mathcal{M}_K$ and performing a "retraction" $\mathcal{R}$ of the iterative update onto $\mathcal{M}_K$.

A problem of interest in the search for finite-time singularity formation in (1) on periodic domain Ω is to study the maximum possible growth of enstrophy within time window T on the constraint manifold

$$\mathcal{M}_K = \left\{ \phi \in H^1(\Omega) \,\middle|\, K = \mathcal{E}_0 \right\} \ . \tag{19}$$

Where the objective functional $\mathcal{J}_T : H^1(\Omega) \to \mathbb{R}$ for this problem can be defined as (6), we seek to solve the optimization problem

$$\widetilde{\varphi} = \underset{\varphi \in \mathcal{M}_K}{\operatorname{argmax}} \, \mathcal{J}_T(\varphi) \ . \tag{20}$$

A numerical solution for Problem (20) can be determined by an iterative process assuming

$$\widetilde{\varphi} = \lim_{n \to \infty} \varphi^{(n)} \ . \tag{21}$$

3.2 Adjoint-Based Optimization

The solution to the optimization problem (5), in accordance with [4] and [3], is formulated in the continuous setting first before discretizing to solve the problem numerically (an "optimize-then-discretize" approach [7]). The augmented objective functional with respect to initial data φ, with Lagrange multiplier $\lambda \in \mathbb{R}$, is defined as

$$\mathcal{J}_\lambda(\phi) = \mathcal{J}_T(\varphi) + \lambda \left(\frac{1}{2} \left\| \frac{d\varphi}{dx} \right\|_{L_2}^2 - \mathcal{E}_0 \right) \ , \tag{22}$$

from which the desired local maximizer $\widetilde{\varphi}$ is characterized by the first-order optimality condition

$$\mathcal{J}_\lambda'(\tilde{\varphi};\varphi') = \mathcal{J}_T'(\tilde{\varphi};\varphi') + \lambda \int_\Omega \frac{d\tilde{\varphi}}{dx}\frac{d\varphi'}{dx} dx \quad \forall \varphi' \in H^1(\Omega) \ , \tag{23}$$

where an arbitrary perturbation direction $\varphi' \in H^1(\Omega)$ allows definition of the Gâteaux derivative $\mathcal{J}_T'(\varphi;\varphi')$ as

$$\mathcal{J}_T'(\varphi;\varphi') = \frac{d}{d\varepsilon} \mathcal{J}_T(\varphi + \varepsilon\varphi')|_{\varepsilon=0} \ . \tag{24}$$

The Sobolev gradient of the objective functional $\nabla \mathcal{J}_T$, defined relative to the Gâteaux derivative via the Riesz Representation theorem as

$$\mathcal{J}_T'(\varphi;\varphi') = \langle \nabla \mathcal{J}_T, \phi' \rangle_{H^1(\Omega)} \tag{25}$$

represents the infinite-dimensional sensitivity of (6) to perturbation direction $\varphi' \in H^1(\Omega)$. This gradient can be obtained by solving the periodic boundary-value problem

$$\nabla \mathcal{J}_T - \ell^2 \frac{d^2}{dx^2} \nabla \mathcal{J}_T = \phi^*|_{t=0} + \frac{d^2}{dx^2}\varphi \quad \in \Omega \ , \tag{26}$$

where ℓ is a length-scale ("regularization") parameter and $\phi^*(t,x)$ solves the periodic adjoint system [18][20]

$$\begin{cases} \partial_t \phi^* + \frac{1}{2}\partial_x (\phi^*)^2 - \nu \partial_{xx}\phi^* = 0 & \in (0,T] \times \Omega \\ \phi^*(T,x) = -\frac{d^2}{dx^2}\phi(T,x) & \in \Omega \end{cases} \quad . \tag{27}$$

3.3 Riemannian Conjugate Gradient

Seeking a local maximizer, we employ the Riemannian conjugate gradient algorithm (Absil et al., 2008 [1]), illustrated in Figure 1. The following derivation holds for any Hilbert space X, but we specifically write $X = H^1$. This momentum-based gradient ascent method is a modification of the nonlinear conjugate gradient algorithm (Nocedal and Wright, 1999 [14]) by:

1. projection $\mathcal{P}_{\mathcal{T}_{\varphi^{(n)}}\mathcal{M}_K} : H^1(\Omega) \to \mathcal{T}_{\varphi^{(n)}}\mathcal{M}_K$ of the gradient of the objective functional $\nabla \mathcal{J}_T(\varphi^{(n)})$ onto the tangent subspace to manifold $\mathcal{M}_K$;
2. vector transport $\mathcal{V}_{\mathcal{T}_{\varphi^{(n)}}\mathcal{M}_K} : \mathcal{T}_{\varphi^{(n-1)}}\mathcal{M}_K \to \mathcal{T}_{\varphi^{(n)}}\mathcal{M}_K$ of the previous search direction $d^{(n-1)}$, if it exists (i.e. excluding the first iteration), via computation of a momentum term $\beta^{(n)} \in \mathbb{R}$ using the Polak-Ribière approach;
3. retraction $\mathcal{R} : H^1(\Omega) \to \mathcal{M}_K$ of the updated direction $\varphi^{(n+1)}$, via computation of an optimal step-size $\tau^{(n)} \in \mathbb{R}$ solved by a variant of Brent's algorithm (Press et al., 1986 [17]).

A summary of the sequential steps in one iteration of the Riemannian conjugate gradient algorithm:

$$\beta^{(n)} = \frac{\left\langle \mathcal{P}_{\mathcal{T}_{\varphi^{(n)}}\mathcal{M}_K} \nabla \mathcal{J}_T(\varphi^{(n)}), \mathcal{P}_{\mathcal{T}_{\varphi^{(n)}}\mathcal{M}_K} \nabla \mathcal{J}_T(\varphi^{(n)}) - \mathcal{V}_{\mathcal{T}_{\varphi^{(n)}}} \mathcal{P}_{\mathcal{T}_{\varphi^{(n-1)}}\mathcal{M}_K} \nabla \mathcal{J}_T(\varphi^{(n-1)}) \right\rangle_{H^1}}{\left\| \mathcal{P}_{\mathcal{T}_{\varphi^{(n-1)}}\mathcal{M}_K} \nabla \mathcal{J}_T(\varphi^{(n-1)}) \right\|_{H^1}^2} \ , \tag{28}$$

$$d^{(n)} = \mathcal{P}_{\mathcal{T}_{\varphi^{(n)}}\mathcal{M}_K} \nabla \mathcal{J}_T(\varphi^{(n)}) + \beta^{(n)} \mathcal{V}_{\mathcal{T}_{\varphi^{(n)}}\mathcal{M}_K} d^{(n-1)} \ , \tag{29}$$

$$\tau^{(n)} = \underset{\tau>0}{\operatorname{argmax}} \left[\mathcal{J}_T \left(\mathcal{R} \left(\varphi^{(n)} + \tau d^{(n)} \right) \right) \right] \ , \tag{30}$$

$$\varphi^{(n+1)} = \mathcal{R}\left(\varphi^{(n)} + \tau^{(n)} d^{(n)} \right) \ . \tag{31}$$

Let ψ be a normal vector to $\mathcal{T}_{\varphi^{(n)}}\mathcal{M}_K$, denoted $\psi \in (\mathcal{T}_{\varphi^{(n)}}\mathcal{M}_K)^\perp$. Then it holds by elementary linear algebra theory that the projection of $\nabla \mathcal{J}_T(\varphi^{(n)})$ onto $\mathcal{T}_{\varphi^{(n)}}\mathcal{M}_K$ is

$$\mathcal{P}^{(n)} := \mathcal{P}_{\mathcal{T}_{\varphi^{(n)}}\mathcal{M}_K} \nabla \mathcal{J}_T(\varphi^{(n)}) = \nabla \mathcal{J}_T(\varphi^{(n)}) - \frac{\langle \psi, \nabla \mathcal{J}_T(\varphi^{(n)}) \rangle_{H^1}}{\|\psi\|_{H^1}^2} \psi \ . \tag{32}$$

The retraction operator $\mathcal{R}$ "normalizes" the iterative update onto the manifold $\mathcal{M}_K$ during the search for optimal step-size $\tau^{(n)}$ in (30), and can be defined for $\varphi^{(n)} + \tau^{(n)} d^{(n)} = \eta^{(n+1)} \in H^1(\Omega)$ as

$$\mathcal{R}^{(n)} := \mathcal{R}\left(\eta^{(n+1)}\right) = \frac{\left\|\varphi^{(n)}\right\|_{H^1}}{\left\|\eta^{(n+1)}\right\|_{H^1}} \eta^{(n+1)} . \tag{33}$$

By setting the momentum term $\beta^{(n)} = 0$, the result in (31) becomes the "Riemannian gradient" method, i.e. steepest ascent on a Riemannian manifold. Using momentum, however, vector transport from $\mathcal{T}_{\varphi^{(n-1)}}\mathcal{M}_K$ to $\mathcal{T}_{\varphi^{(n)}}\mathcal{M}_K$ can be computed via "differentiated retraction" (Absil et al., 2008 [1]). Taking $\varphi^{(n-1)} \in \mathcal{M}_K$ and $\tau^{(n-1)} d^{(n-1)} \in \mathcal{T}_{\varphi^{(n-1)}}\mathcal{M}_K$ to define the vector for retraction, we compute the derivative of $\mathcal{R}\left(\varphi^{(n-1)} + \tau^{(n-1)} d^{(n-1)}\right) = \mathcal{R}\left(\eta^{(n)}\right)$ in the direction of $d^{(n-1)} \in \mathcal{T}_{\varphi^{(n-1)}}\mathcal{M}_K$: where

$$\nabla_{d^{(n-1)}}\left(\eta^{(n)}\right) = \frac{d}{dt}\left[\eta + t d^{(n-1)}\right]_{t=0} = d^{(n-1)} , \tag{34}$$

$$\nabla_{d^{(n-1)}}\left\|\eta^{(n)}\right\|_{H^1}^2 = \frac{d}{dt}\left[\left\|\eta^{(n)}\right\|_{H^1}^2 + \left\|t d^{(n-1)}\right\|_{H^1}^2 + 2\langle \eta^{(n)}, t d^{(n-1)}\rangle_{H^1}\right]_{t=0} = 2\langle \eta^{(n)}, d^{(n-1)}\rangle_{H^1} , \tag{35}$$

$$\nabla_{d^{(n-1)}}\left(\frac{1}{\left\|\eta^{(n)}\right\|_{H^1}}\right) = -\frac{\nabla_{d^{(n-1)}}\left\|\eta^{(n)}\right\|_{H^1}}{\left\|\eta^{(n)}\right\|_{H^1}^2} = -\frac{\langle \eta^{(n)}, d^{(n-1)}\rangle_{H^1}}{\left\|\eta^{(n)}\right\|_{H^1}^3} , \tag{36}$$

it follows that vector transport via differentiated retraction can be expressed as

$$\begin{aligned}\mathcal{V}^{(n)} := \mathcal{V}_{\mathcal{T}_{\varphi^{(n)}}\mathcal{M}_K} d^{(n-1)} &= \nabla_{d^{(n-1)}}\mathcal{R}\left(\eta^{(n)}\right) \\ &= \left\|\varphi^{(n)}\right\|_{H^1} \frac{d}{dt}\left[\frac{\eta^{(n)}}{\left\|\eta^{(n)}\right\|_{H^1}}\right]_{t=0} \\ &= \left\|\varphi^{(n)}\right\|_{H^1}\left(\frac{\nabla_{d^{(n-1)}}\left(\eta^{(n)}\right)}{\left\|\eta^{(n)}\right\|_{H^1}} + \eta^{(n)}\nabla_{d^{(n-1)}}\left(\frac{1}{\left\|\eta^{(n)}\right\|_{H^1}}\right)\right) \\ &= \frac{\left\|\varphi^{(n)}\right\|_{H^1}}{\left\|\eta^{(n)}\right\|_{H^1}}\left(d^{(n-1)} - \frac{\langle \eta^{(n)}, d^{(n-1)}\rangle_{H^1}}{\left\|\eta^{(n)}\right\|_{H^1}^2}\eta^{(n)}\right) .\end{aligned} \tag{37}$$

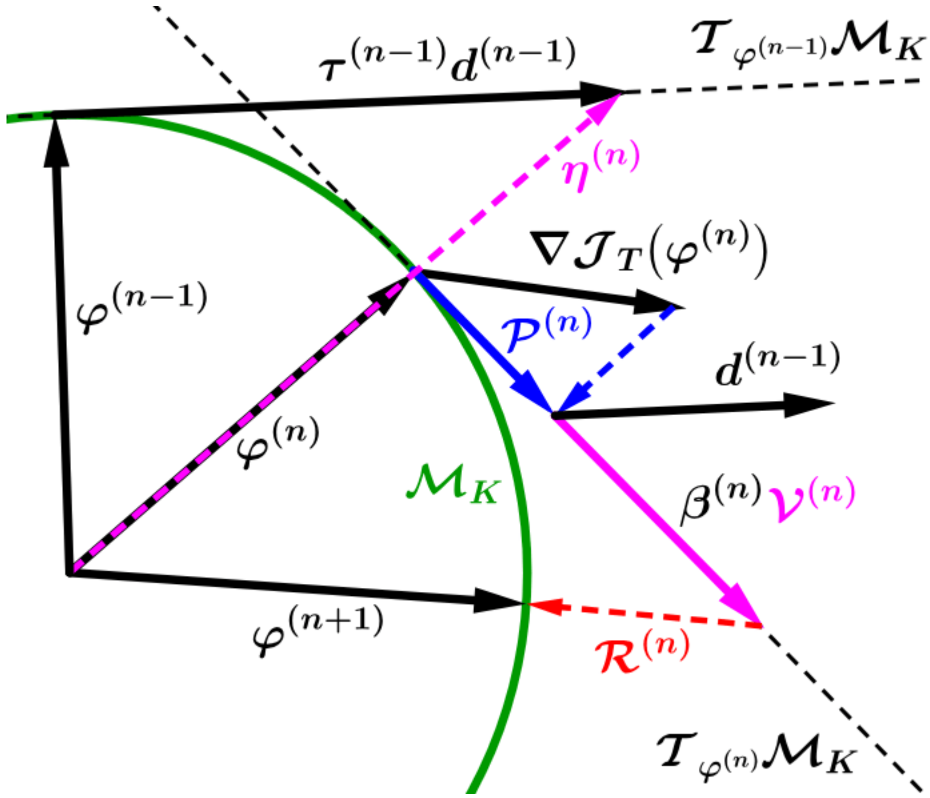

Figure 1: Schematic of Riemannian Conjugate Gradient Method

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